Flymedia

SAAS PROJECT MANAGEMENT PLATFORM

Project Documentation & Workflow Guide

Architecture · User Roles · Project Lifecycle · API Reference

VERSION	DATE	STATUS
2.0	March 2026	Production

CHAPTER 01

Project Overview

Flymedia is a multi-tenant SaaS project management platform designed for agencies and media organizations. It enables structured collaboration between *clients*, *project teams*, and *platform administrators* from a single unified interface.

The platform manages the entire project lifecycle — from a client submitting a project request, through organizational approval, team assignment, task execution, document sharing, and final delivery — all within a role-aware, secure environment.

> **Core Philosophy:** Every piece of functionality is gated by role and tenant, ensuring organizations work in isolated, secure workspaces while superadmins maintain visibility across the entire platform.

Key Capabilities

- **Multi-Tenancy:** Each registered organization (tenant) is isolated. Data, projects, and users belong to a single tenant.
- **Role-Based Access Control (RBAC):** Six distinct roles each with precisely defined permissions.
- **Client-Initiated Projects:** Clients can submit project requests to specific organizations; admins approve and launch them.
- **Kanban & List Views:** Drag-and-drop board plus a sortable, filterable list view for task management.
- **Real-Time Chat:** Built-in messaging with Socket.IO, file sharing, and per-project channels.
- **Document Management:** Clients and team members upload files directly to tasks and projects.
- **Password Recovery:** OTP-based forgot password flow via email.

Platform Users (Roles)

Super Admin

Full platform access. Manages all tenants, views cross-organization data, and monitors platform health.

Admin

Organization administrator. Manages employees, projects, and approves client project requests within their tenant.

Manager

Oversees specific projects. Assigns tasks to employees and tracks progress through to completion.

Employee

Executes assigned tasks. Uploads deliverables and logs progress updates to the project board.

Client

Submits project requests to organizations. Monitors progress and views uploaded documents for their projects.

System

Automated system processes including email notifications and scheduled status transitions.

CHAPTER 02

System Architecture

Flymedia is a decoupled full-stack application with a REST API backend and a React-based frontend. Real-time features are layered on using WebSockets.

Technology Stack

LAYER	TECHNOLOGY	RESPONSIBILITY	CATEGORY
Frontend Framework	Next.js 14 (App Router)	Server-side rendering, routing, page structure	Core
UI	React + Vanilla CSS	Component rendering, state management, styling	Core
Backend API	Node.js + Express.js	REST API, business logic, middleware	Server
Database	MongoDB + Mongoose	Data persistence, schema validation, querying	Server
Authentication	JWT + HTTP Cookies	Stateless auth, role-based access, session persistence	Server
Real-Time	Socket.IO	Live chat, message delivery, user presence	Real-Time
File Uploads	Multer (disk storage)	Task attachments, project documents, chat media	Server
Email	Nodemailer + Gmail SMTP	OTP delivery, password reset, notifications	External
HTTP Client	Axios	Frontend API communication with interceptors	Core
Drag & Drop	@dnd-kit	Kanban board task reordering across columns	Core

Application Structure

Backend (/backend)

- **server.js** — Express app entry point, middleware, socket setup, route mounting
- **models/** — Mongoose schemas: User, Tenant, Project, Task, Message
- **controllers/** — Business logic: auth, projects, tasks, employees, tenants, dashboard
- **routes/** — API route definitions with middleware chains
- **middleware/** — Auth guard (JWT verify), role authorization, Multer file upload
- **utils/** — Email sender utility (Nodemailer)
- **public/uploads/** — Uploaded files organized by user and type

Frontend (/web)

- **app/** — Next.js App Router pages (login, register, dashboard, projects, tasks, messages)
- **components/** — Reusable UI (Button, Input, Dialog, Badge, KanbanBoard, TaskDetailModal, Sidebar, Chat components)
- **context/** — React Contexts: AuthContext (user state), SocketContext (real-time)
- **lib/api.js** — Axios instance with base URL and credential configuration

Data Models

MODEL	KEY FIELDS	RELATIONSHIPS
User	name, email, password (hashed), role, tenant, resetPasswordOTP	Belongs to Tenant
Tenant	name, domain, createdAt	Has many Users, Projects
Project	name, description, status, client, manager, team[], tenant, files[]	Belongs to Tenant; refs User (client/manager)
Task	title, description, status, priority, dueDate, assignedTo, attachments[], activityLog[]	Belongs to Project and Tenant

MODEL	KEY FIELDS	RELATIONSHIPS
Message	senderId, receiverId, message, projectId, attachments[]	Refs User, optionally Project

CHAPTER 03

Project Workflow

The complete lifecycle of a project on Flymedia flows through well-defined stages, each involving specific user roles and system actions.

Project Status Lifecycle

Requested → Planned → In Progress → Review → Completed or Rejected

End-to-End Workflow

1

Client Registration & Login

A client registers on the platform, automatically creating or joining a tenant workspace. They log in to receive a JWT stored in a secure cookie for session persistence across page refreshes.

2

Client Submits Project Request

The client navigates to *Request Project*, fills in project details (name, description, timeline), selects a target organization from the list of registered tenants, and optionally uploads reference files. The project is created with status

Requested

.

3

Organization Reviews the Request

The tenant's Admin or Manager sees the incoming request in their Projects list. They can review the details, attached documents, and the client's contact information before making a decision.

4

Admin Approves or Rejects

If approved, the Admin clicks *Approve Request*, which sets the project status to

Planned

and optionally assigns a manager. If rejected, the client is notified and the project is marked

Rejected

.

5

Manager Assigns Tasks

The assigned Manager opens the project's Kanban board, creates tasks with titles, descriptions, priorities (Low/Medium/High), and due dates. Each task is assigned to an eligible team employee. The project status moves to

In Progress

.

6

Employees Execute & Upload

Employees see only their assigned tasks. They update task status by dragging cards between Kanban columns (To Do → In Progress → Review → Done). They can open a task to upload deliverable files and view the activity log.

7

Manager Reviews Work

The Manager reviews tasks in the

Review

column, provides feedback via chat (using the project-scoped messaging channel), requests revisions, or moves the task to

Done

.

8

Client Monitors Progress

Throughout the project, the client can log in to view the current task statuses in list or board view, download uploaded documents, and communicate directly with the project team using the built-in chat.

9

Project Completed

When all tasks are marked done and final deliverables are uploaded, the Manager marks the project as

Completed

. All attached documents remain accessible to the client for download.

CHAPTER 04

Authentication & Security

Registration Flow

When a new organization registers:

1. User submits name, email, password, and company (tenant) name + domain.
2. Backend looks up the tenant by name and domain. If not found, creates a new **Tenant** document.
3. Creates the **User** document (role defaults to `admin`) linked to the tenant.
4. Returns a signed **JWT** stored in an HTTP cookie (30-day expiry).

Login & Session Persistence

- On login, the backend issues a JWT containing `{ id, role, tenant }`.
- On each page load, the frontend calls `GET /api/auth/me` using the cookie to re-hydrate the user session — preventing logout on refresh.
- If the server returns `401 Unauthorized`, the user is logged out and redirected to login.

Role-Based Authorization

Every protected API route uses two middleware layers:

- **protect** — Verifies the JWT, populates `req.user` with id, role, and tenant.
- **authorize(...roles)** — Checks the user's role against an allowlist. Returns `403 Forbidden` if not permitted.

Forgot Password Flow

1 **Request OTP**

User submits their email on the *Forgot Password* page. Server generates a 6-digit OTP, saves it with a 10-minute expiry using `User.updateOne()` (bypassing password hash hook), and sends it via Gmail SMTP.

2 **Verify & Reset**

User enters the OTP and new password on the *Reset Password* page. Server queries by email + OTP + non-expired date. On match, saves the new password (triggering bcrypt hash), clears OTP fields, and signs a new JWT.

3 **Auto Login**

After a successful reset, the user is automatically logged in and redirected to the dashboard.

Security Highlights

Tenant Isolation: All queries filter by `req.user.tenant` to prevent cross-organization data access. Superadmins explicitly bypass this filter.

Password Security: Passwords are hashed using **bcryptjs** (salt rounds: 10) via a Mongoose pre-save hook. Plaintext passwords are never stored or logged.

File Access: Uploaded files are served from `/public/uploads/` by the Express static middleware, accessible by anyone with the direct URL. Sensitive projects should use signed URLs for production.

API Endpoints Summary

METHOD	ENDPOINT	ACCESS	DESCRIPTION
POST	/api/auth/register	Public	Register new organization + admin user

METHOD	ENDPOINT	ACCESS	DESCRIPTION
POST	/api/auth/login	Public	Login, returns JWT in cookie
GET	/api/auth/me	Auth	Get current user (session check)
POST	/api/auth/forgotpassword	Public	Send OTP to email
POST	/api/auth/resetpassword	Public	Verify OTP and set new password
GET	/api/projects	Auth	List projects (role-filtered)
POST	/api/projects	Auth	Create / request project
PUT	/api/projects/:id/approve	Admin/Manager	Approve a requested project
GET	/api/tasks	Auth	List tasks (project/role-filtered)
POST	/api/tasks	Non-client	Create new task
PUT	/api/tasks/:id	Auth	Update task status, priority, etc.
POST	/api/tasks/:id/attachments	Auth	Upload files to task
GET	/api/employees	Auth	List users (excl. superadmin & client)
GET	/api/tenants	Auth	List all organizations
GET	/api/tenants/:id	SuperAdmin	Tenant details + linked projects
GET	/api/messages/:receiverId	Auth	Get conversation messages
POST	/api/messages/send/:receiverId	Auth	Send message (with optional file)

CHAPTER 05

Feature Deep Dives

Kanban Board & List View

The project detail page provides two task visualization modes switchable by the view toggle in the header:

- **Kanban Board:** Drag-and-drop columns (To Do / In Progress / Review / Done) powered by `@dnd-kit`. Moving a card makes a `PUT /api/tasks/:id` call to persist the new status.
- **List View:** A sortable table showing all tasks. Columns (Title, Status, Priority, Due Date) are all sortable by clicking the column header. Priority items are color-coded (red=high, yellow=medium, blue=low).

> **Task Sorting:** The list view sorts client-side using a comparison function that maps priority strings (high/medium/low) to numeric values for correct ordering.

Real-Time Chat

The messaging system connects users via WebSocket (Socket.IO) with the following features:

- **Direct Messages:** User-to-user conversation threads.
- **Project-Scoped Chat:** Conversations can be tagged with a `projectId` to keep discussions organized per project.
- **File Sharing:** Users can attach images and documents to messages.
- **Message Forwarding:** Any message can be forwarded to another user via the ForwardModal.
- **Live Delivery:** New messages are pushed via `newMessage` socket events without polling.

Document Management

Files can be attached at two levels in the platform:

- **Project Files:** Uploaded when creating or approving a project (client reference documents, briefs).

- **Task Attachments:** Uploaded within the Task Detail Modal by any authorized user. Each attachment records the uploader name, upload date, and file type (image/document). Files are served as direct links from the server's static directory.

Dashboard Analytics

Each role sees a personalized dashboard with relevant statistics:

- **SuperAdmin:** Total tenants, users across the platform, recent activity.
- **Admin/Manager:** Active projects, in-progress tasks, team workload, recent project requests.
- **Employee:** Tasks assigned to them, their completion rate.
- **Client:** Their submitted project requests and current statuses.

Tenant Management (SuperAdmin)

The SuperAdmin has exclusive access to the Tenants section:

- Lists all registered organizations with domain links.
- Clicking *View Details* on a tenant card shows a detail page with the organization's project history, stats, and direct links to individual project boards.
- Can create new tenant organizations and provision admin users.

Sidebar Navigation (Role-Aware)

The dashboard sidebar dynamically renders navigation links based on the logged-in user's role:

LINK	VISIBLE TO
Dashboard	All roles
Projects	All roles
Request Project	Client only
Employees	SuperAdmin, Admin
Tenants	SuperAdmin, Admin
Tasks	SuperAdmin, Admin, Employee, Manager

LINK	VISIBLE TO
Messages	All roles

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